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HydroPulse-SVR: El Niño-Based KRS Reservoir Inflow Deficit Prediction Using Support Vector Regression

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Abstract

Reservoir inflow in the Cauvery basin depends on monsoon rainfall, which can be influenced by the El Niño–Southern Oscillation (ENSO). This project presents HydroPulse-SVR, a Support Vector Regression (SVR) model that predicts the next-month inflow deficit (%) of the Krishnarajasagara (KRS) reservoir from ENSO indices (ONI, SOI and lagged ONI), catchment rainfall, current and previous reservoir inflow, reservoir storage and seasonal features for 2011–2020. The predicted deficit is converted into a five-level hydropower-risk indicator and presented through an interactive six-page Streamlit dashboard. On the chronologically held-out test months (January 2018 – November 2020), the SVR-RBF model achieves an RMSE of 60.17 and an MAE of 43.94 percentage points with an R^2 of 0.402, better than Linear Regression, Ridge Regression, Random Forest and a Persistence baseline.

Background

El Niño can change monsoon rainfall, rainfall drives reservoir inflow, and lower inflow raises the risk to hydropower water availability. Because the ENSO–monsoon relationship has weakened in recent decades [8], ENSO alone is not enough, so this project combines it with rainfall, reservoir inflow, storage and seasonal information. KRS, in the Cauvery basin of Karnataka, regulates irrigation and municipal drinking water and is the master control reservoir whose discharge dictates head and power generation at the downstream cascade stations: Shivanasamudra (96 MW), Shimsha (17.2 MW) and Mettur Dam (200 MW).

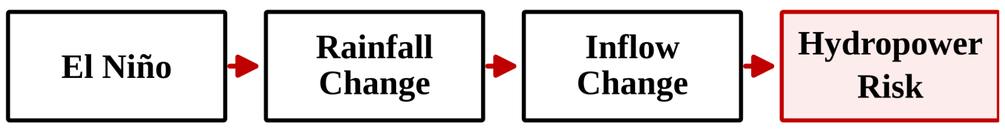


Fig. 1: Chain reaction from El Niño to hydropower risk

Objectives

- To combine ENSO, rainfall and KRS reservoir data into a single dataset.
- To convert the data into a consistent monthly dataset.
- To create climate, hydrological and seasonal features, including lagged ENSO values, previous inflow and a cyclic encoding of the calendar month.
- To train and evaluate an SVR-RBF model using a chronological split and time-series cross-validation.
- To compare SVR with baseline models (Linear Regression, Ridge Regression, Random Forest and Persistence).
- To develop an interactive Streamlit dashboard for visualisation and prediction.
- To represent the predicted inflow deficit as a hydropower-risk indicator that can be understood without Machine Learning knowledge.

Methodology

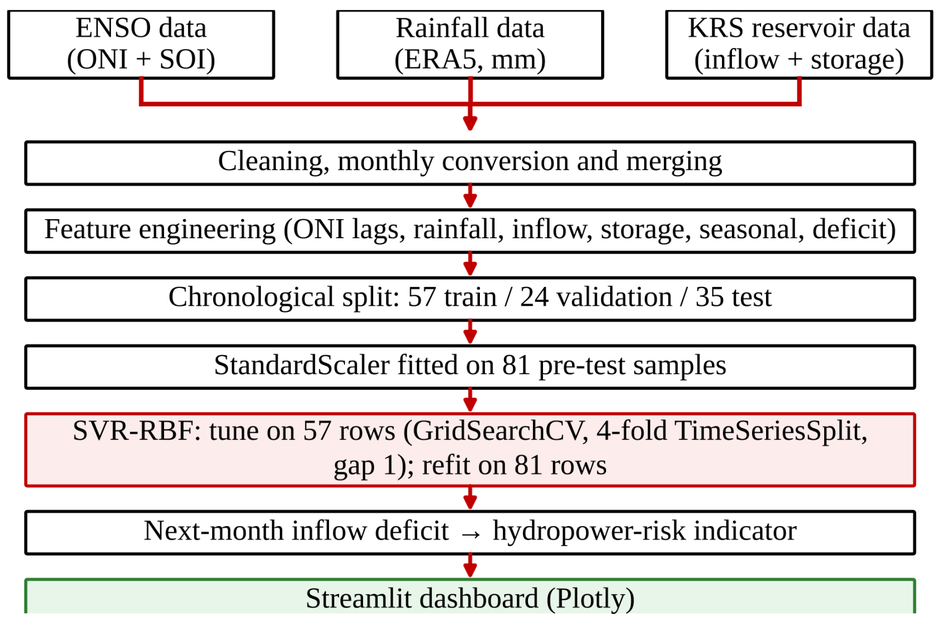


Fig. 2: Project workflow of HydroPulse-SVR

NOAA CPC/ENSOcast [5], World Bank ERA5 rainfall [6] and India-WRIS KRS gauge data [7] cover 2011–2020 (120 months; 116 usable samples; 11 input features). The deficit is $D(t) = 100 \times [N(m) - \text{inflow}(t)] / N(m)$, where $N(m)$ is the mean inflow of calendar month m over 2011–2016. The samples are split chronologically into 57 training (49.1%), 24 validation (20.7%) and 35 test (30.2%) samples; final fitting uses the 81 pre-test rows. The SVR-RBF grid search covers 96 combinations: C = 10, 50, 100, 200; γ = 0.005, 0.01, 0.02, 0.05, 0.1, scale; ε = 0.5, 1, 2, 5. Caveats: scaling precedes cross-validation, normals include 2016, and target clipping uses full-data quantiles.

Risk indicator: below 0% Surplus; 0% to <20% Low; 20% to <40% Moderate; 40% to <60% High; 60% and above Severe.

Result and Discussion

60.17	43.94	0.402	37.1%
RMSE	MAE	R^2	Within ±20 points

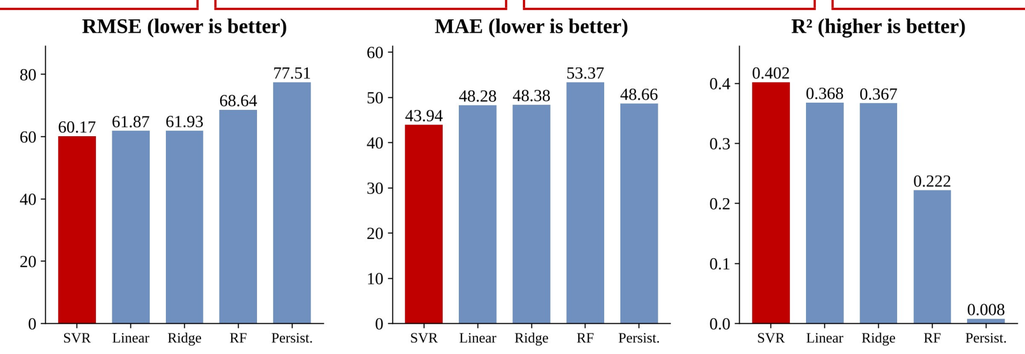


Fig. 3: Test-set comparison of SVR-RBF (red) with baseline models (Jan 2018 – Nov 2020)

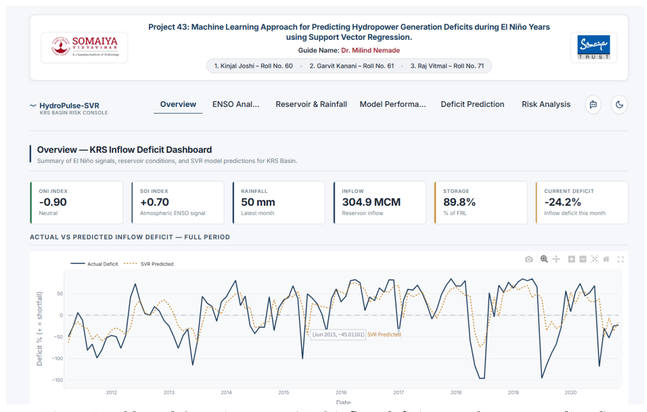


Fig. 4: Dashboard Overview page (KRS inflow deficit, actual versus predicted)

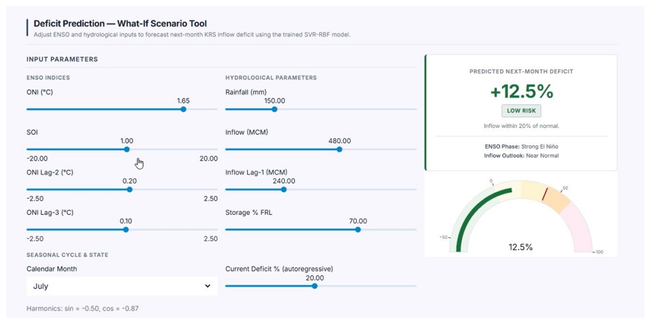


Fig. 5: Deficit Prediction page (what-if scenario tool)

- SVR-RBF leads all baselines on RMSE, MAE and R^2 (60.17, 43.94 and 0.402), explaining about 40.2% of test-set variance.
- Its RMSE is lower by about 2.7%, 12.3% and 22.4% than Linear Regression, Random Forest and Persistence respectively.
- Linear (R^2 0.368) and Ridge (0.367) remain close; Random Forest has R^2 0.222; Persistence has the highest RMSE (77.51) and R^2 0.008.
- Within ±20 percentage points, SVR and Persistence each achieve 37.1% (13 of 35 months), so the gain of SVR comes mainly from reducing large errors.
- Results cover one reservoir and a 35-month test set; ENSO attribution is unverified.

Conclusion

HydroPulse-SVR is a complete pipeline that combines ENSO, rainfall and KRS reservoir data to predict the next-month inflow deficit and present it as a hydropower-risk indicator in an interactive Streamlit dashboard. SVR-RBF gave the best results on all three metrics on the held-out test months. The accuracy is modest (R^2 0.402), so the output should support, not replace, operational data and expert judgement. Future work: more Cauvery-basin reservoirs (Harangi, Kabini, Hemavathi), longer data, actual generation data (MWh/GWh), longer forecast horizons, LSTM models and a strictly leakage-free evaluation.

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